

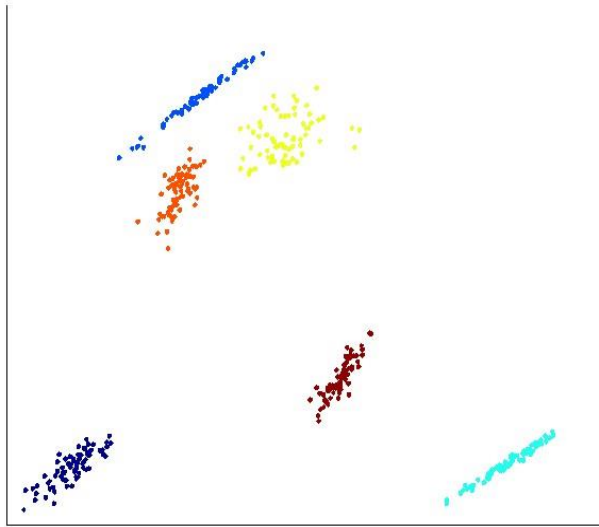
Artificial Intelligence

Lecture 14: Clustering

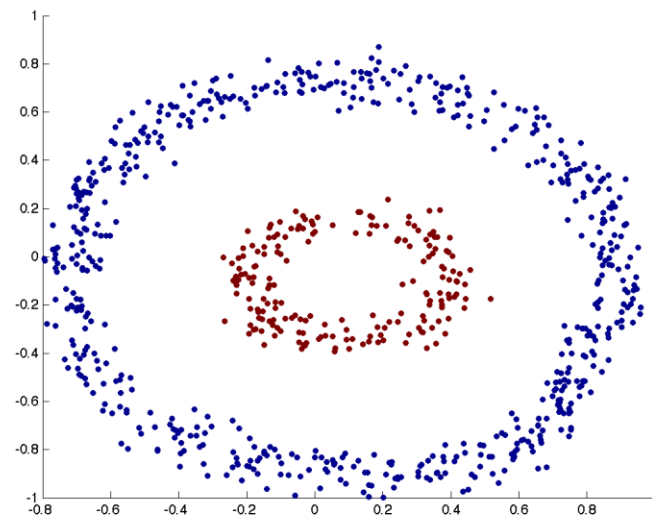
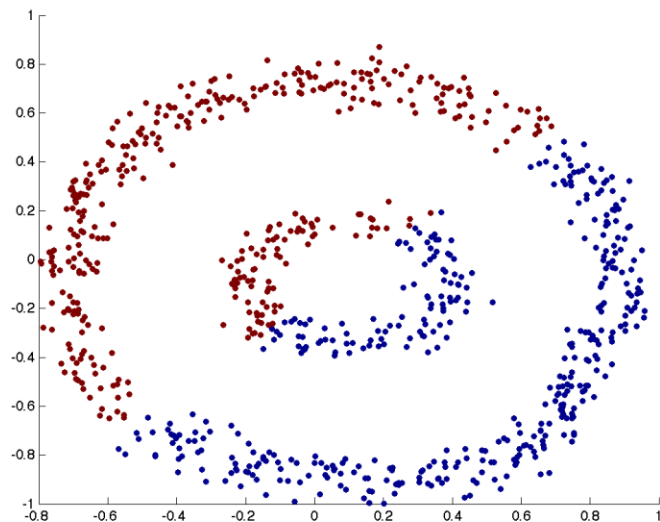
Algorithms

- Partitioning approach:
 - Construct various partitions and then evaluate them by some criterion, e.g., minimizing the sum of square errors
 - Typical methods: **k-means, k-medoids**
- Model-based:
 - A model is hypothesized for each of the clusters and tries to find the best fit of that model to each other
 - Typical methods: **GMM**
- Dimensionality reduction approach
 - First dimensionality reduction, then clustering
 - Typical methods: **Spectral clustering**, Ncut

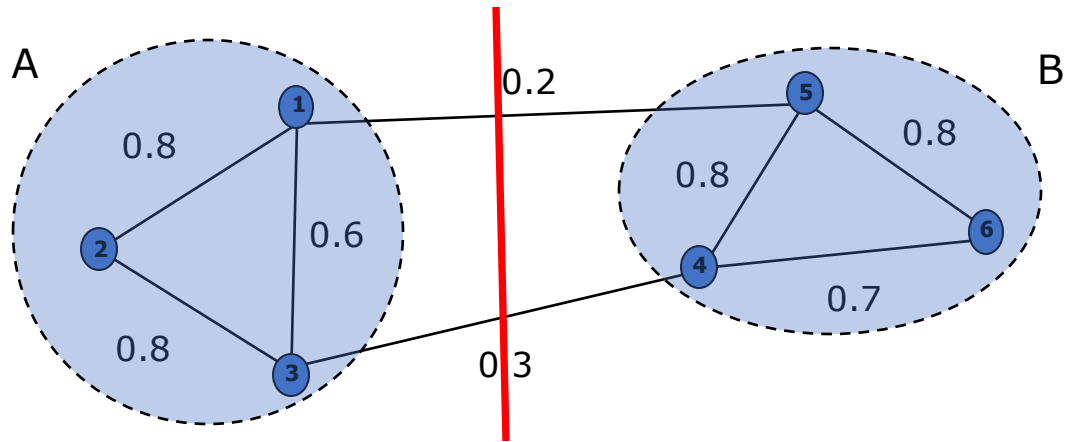
Good clustering – we know it when we see it



An Example

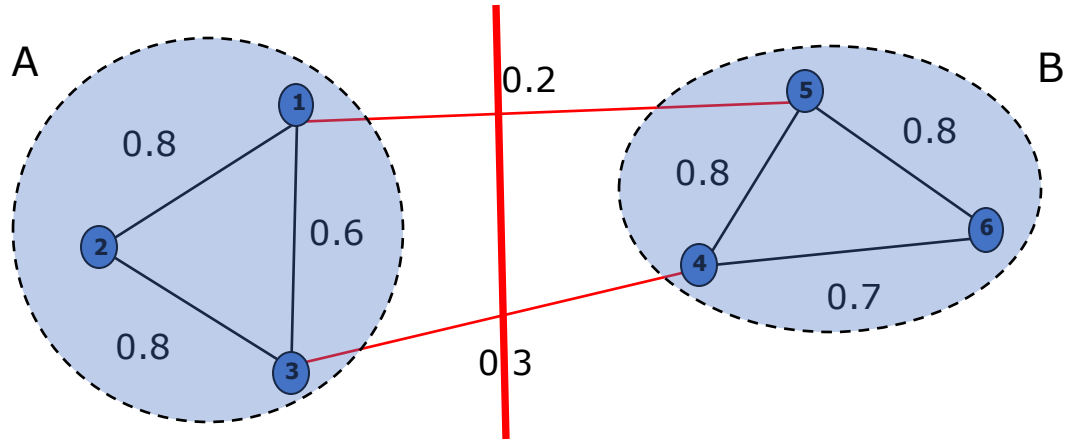


Spectral Clustering



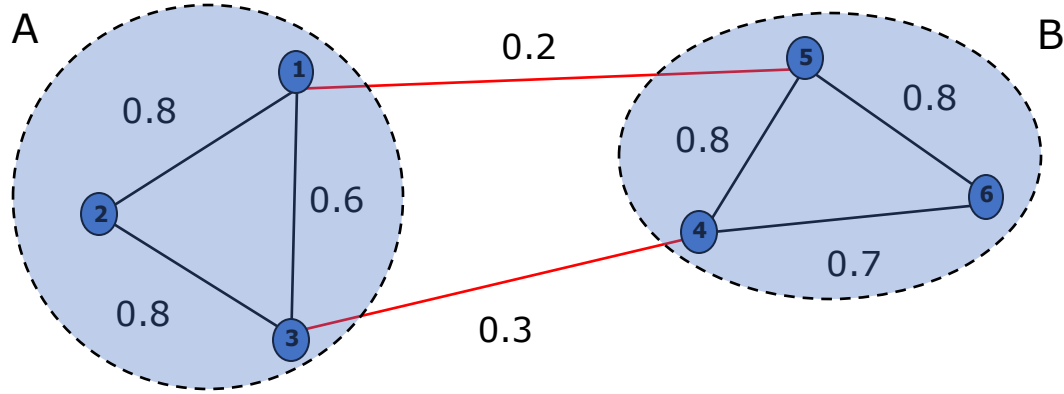
- Represent data points as the vertices V of a graph G .
 - All pairs of vertices are connected by an edge E .
 - Edges have weights W . Large weights mean that the adjacent vertices are very similar; small weights imply dissimilarity.
- Clustering can be viewed as partitioning a similarity graph
 - Divide vertices into two disjoint groups (A,B)

Clustering Objectives



- Traditional definition of a “good” clustering:
 - Points assigned to same cluster should be highly similar.
 - Points assigned to different clusters should be highly dissimilar.
- Apply these objectives to our graph representation
 - Minimize weight of **between-group** connections

Graph Cuts

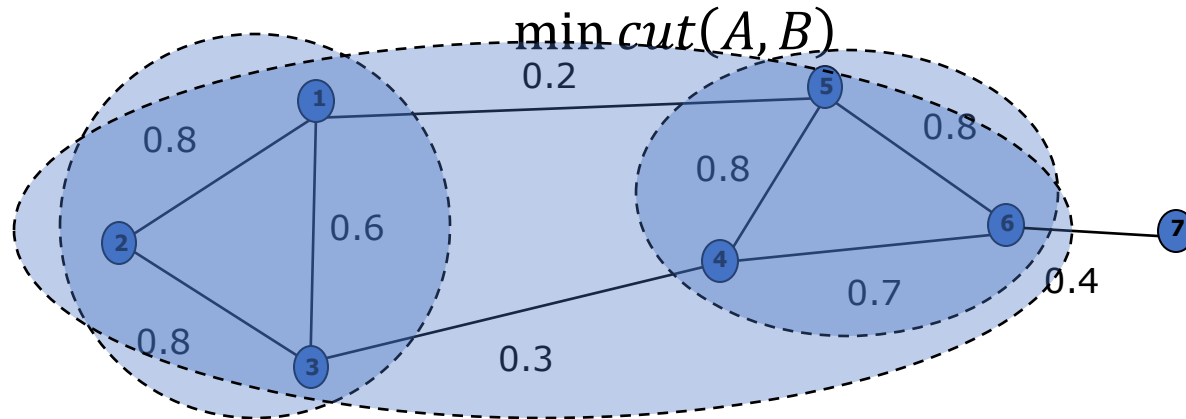


- Express partitioning objectives as a function of the “edge cut” of the partition.
 - Cut:** Set of edges with only one vertex in a group. We want to find the minimal cut between groups. The groups that have the minimal cut would be the partition

$$cut(A, B) = \sum_{i \in A, j \in B} w_{ij}$$

Graph Cut Criteria

- Criterion: Minimum-cut
 - Minimize the weights of connections between groups



- Problem:
 - Only considers the inter-cluster connections
 - Does not consider the intra-cluster density

- ▶ Maximize the weights of connections within groups

$$\max(\text{assoc}(A, A) + \text{assoc}(B, B))$$

- $\text{assoc}(A, A) = \sum_{i \in A, j \in A} w_{ij}$

Graph Cut Criteria

- Criterion: Normalized-cut (Shi & Malik,'97): Normalized Cuts and Image Segmentation

- Consider the connectivity between groups relative to the density of each group.

$$Ncut(A, B) = \frac{cut(A, B)}{assoc(A, V)} + \frac{cut(A, B)}{assoc(B, V)}$$

- Normalize the association between groups.

$$assoc(A, V) = \sum_{i \in A, j \in V} w_{ij}$$

- Produces more balanced partitions

$$\min Ncut(A, B)$$

$$Nassoc(A, B) = \frac{assoc(A, A)}{assoc(A, V)} + \frac{assoc(B, B)}{assoc(B, V)}$$

Graph Cut Criteria

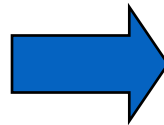
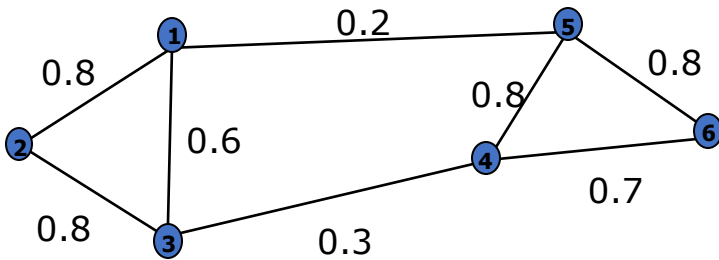
$$cut(A, B) = assoc(A, V) - assoc(A, A)$$

$$cut(A, B) = assoc(B, V) - assoc(B, B)$$

- $Ncut(A, B) = \frac{cut(A, B)}{assoc(A, V)} + \frac{cut(A, B)}{assoc(B, V)}$
- $= \frac{assoc(A, V) - assoc(A, A)}{assoc(A, V)} + \frac{assoc(B, V) - assoc(B, B)}{assoc(B, V)}$
- $= 2 - \left(\frac{assoc(A, A)}{assoc(A, V)} + \frac{assoc(B, B)}{assoc(B, V)} \right) = 2 - Nassoc(A, B)$

Matrix Representation

- Adjacency matrix (W)
 - $n \times n$ matrix
 - w_{ij} : edge weight between vertex x_i and x_j
 - Symmetric matrix



	x_1	x_2	x_3	x_4	x_5	x_6
x_1	0	0.8	0.6	0	0.2	0
x_2	0.8	0	0.8	0	0	0
x_3	0.6	0.8	0	0.3	0	0
x_4	0	0	0.3	0	0.8	0.7
x_5	0.2	0	0	0.8	0	0.8
x_6	0	0	0	0.7	0.8	0

Objective Function of Ncut



$$\mathbf{x} \in [1, -1]^n, x_i = \begin{cases} 1 & i \in A \\ -1 & i \in B \end{cases} \quad d_i = \sum_j w_{ij}$$

$$\begin{aligned} Ncut(A, B) &= \frac{cut(A, B)}{assoc(A, V)} + \frac{cut(A, B)}{assoc(B, V)} \\ &= \frac{\sum_{x_i > 0, x_j < 0} -w_{ij} x_i x_j}{\sum_{x_i > 0} d_i} + \frac{\sum_{x_i < 0, x_j > 0} -w_{ij} x_i x_j}{\sum_{x_i < 0} d_i} \end{aligned}$$

$$W \in R^{n \times n} \quad D \in R^{n \times n} \quad \mathbf{x} \in [1, -1]^n \quad \mathbf{1} \in [1]^n \quad k = \frac{\sum_{x_i > 0} d_i}{\sum_i d_i}$$

$$4 Ncut(A, B) = \frac{(\mathbf{1} + \mathbf{x})^T (D - W) (\mathbf{1} + \mathbf{x})}{k \mathbf{1}^T D \mathbf{1}} + \frac{(\mathbf{1} - \mathbf{x})^T (D - W) (\mathbf{1} - \mathbf{x})}{(1 - k) \mathbf{1}^T D \mathbf{1}}$$

$$b = \frac{k}{1 - k}$$

$$= \frac{[(\mathbf{1} + \mathbf{x}) - b(\mathbf{1} - \mathbf{x})]^T (D - W) [(\mathbf{1} + \mathbf{x}) - b(\mathbf{1} - \mathbf{x})]}{b \mathbf{1}^T D \mathbf{1}}$$

Objective Function of Ncut



$$\mathbf{y} = (\mathbf{1} + \mathbf{x}) - b(\mathbf{1} - \mathbf{x}) \quad k = \frac{\sum_{x_i > 0} d_i}{\sum_i d_i} \quad b = \frac{k}{1 - k} = \frac{\sum_{x_i > 0} d_i}{\sum_{x_i < 0} d_i}$$

$$\mathbf{y}^T D \mathbf{1} = 2 \sum_{x_i > 0} d_i - 2b \sum_{x_i < 0} d_i = 0$$

$$\begin{aligned} \mathbf{y}^T D \mathbf{y} &= 4 \sum_{x_i > 0} d_i + 4b^2 \sum_{x_i < 0} d_i = 4 \left(b \sum_{x_i < 0} d_i + b^2 \sum_{x_i < 0} d_i \right) \\ &= 4b \left(\sum_{x_i < 0} d_i + b \sum_{x_i < 0} d_i \right) = 4b \mathbf{1}^T D \mathbf{1} \end{aligned}$$

$$\begin{aligned} \min_{\mathbf{x}} Ncut(\mathbf{x}) &= \min_{\mathbf{y}} \frac{\mathbf{y}^T (D - W) \mathbf{y}}{\mathbf{y}^T D \mathbf{y}} \\ \text{s.t. } \mathbf{y} &\in [2 - 2b, -2b]^n, \quad \mathbf{y}^T D \mathbf{1} = 0 \end{aligned}$$

- NP-hard!

Rayleigh quotient

- Relaxation:

$$\min_{\mathbf{y}} \frac{\mathbf{y}^T (D - W) \mathbf{y}}{\mathbf{y}^T D \mathbf{y}}, \mathbf{y} \in \mathcal{R}^n, \mathbf{y}^T D \mathbf{1} = 0$$

- $L \equiv D - W$

$$\min_{\mathbf{y}} \frac{\mathbf{y}^T L \mathbf{y}}{\mathbf{y}^T D \mathbf{y}}, \mathbf{y} \in \mathcal{R}^n, \mathbf{y}^T D \mathbf{1} = 0$$

Rayleigh quotient

$$\max_x \frac{x^T A x}{x^T B x} \quad \longrightarrow \quad \max_x x^T A x \quad s.t. \quad x^T B x = 1$$

Lagrangian Function

$$L(x) = x^T A x + \lambda(x^T B x - 1)$$

Taking the derivative with respect to x

$$\frac{\partial L(x)}{\partial x} = 0 \quad \longrightarrow \quad (A + A^T)x + \lambda(B + B^T)x = 0$$

If A and B are symmetric

$$Ax = \kappa Bx, \kappa = -\lambda$$

General Eigen Decomposition

Generalized Eigen-problem

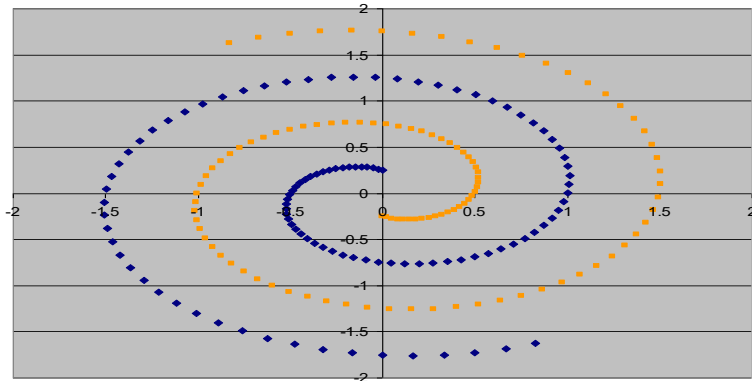
$$\min_{\mathbf{y}} \frac{\mathbf{y}^T (D - W) \mathbf{y}}{\mathbf{y}^T D \mathbf{y}}, \mathbf{y} \in \mathcal{R}^n, \mathbf{y}^T D \mathbf{1} = 0$$

$$(D - W) \mathbf{y} = \lambda D \mathbf{y}$$

$$\begin{aligned} (D - W) \mathbf{y} &= \lambda D^{\frac{1}{2}} D^{\frac{1}{2}} \mathbf{y} \\ D^{-\frac{1}{2}} (D - W) D^{-\frac{1}{2}} D^{\frac{1}{2}} \mathbf{y} &= \lambda D^{\frac{1}{2}} \mathbf{y} \\ D^{-\frac{1}{2}} (D - W) D^{-\frac{1}{2}} \mathbf{z} &= \lambda \mathbf{z} \end{aligned}$$

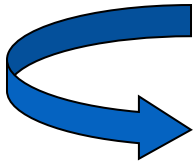
- Eigenvector corresponding to the **smallest** eigenvalue.
- Vector **1** is the eigenvector corresponding to the eigenvalue 0.
- The eigenvector corresponding to the **2nd** small eigenvalue.

Spectral Clustering Example – 2 Spirals

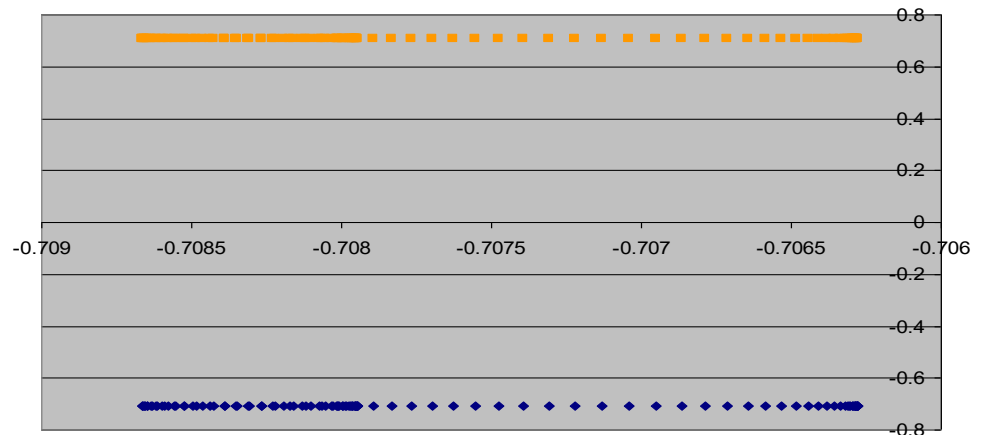


Dataset exhibits complex cluster shapes

⇒ K-means performs very poorly in this space due bias toward dense spherical clusters.



In the embedded space given by two leading eigenvectors, clusters are trivial to separate.



$$K > 2$$

- Perform Ncut recursively.
- Use more than one eigenvectors.
 - Suppose $\mathbf{y}_1, \mathbf{y}_2, \dots, \mathbf{y}_k$ are the first k eigenvectors corresponding to the smallest eigenvalues, let
$$Y = [\mathbf{y}_1, \mathbf{y}_2, \dots, \mathbf{y}_k] \in R^{n \times k}$$
 - Each row vector of Y is a k dimensional representation of the original data point.
 - Performing kmeans.

Spectral Clustering Algorithm

1. Graph construction

- Heat kernel $w_{ij} = \exp\left\{-\frac{\|x_i - x_j\|}{2\sigma^2}\right\}$
- k -nearest neighbor graph

2. Eigen-problem

- Compute eigenvalues and eigenvectors of the matrix L
- Map each point to a lower-dimensional representation based on one or more eigenvectors.

3. Conventional clustering schemes, e.g. K-Means

- Assign points to two or more clusters, based on the new representation.